

RESEARCH INTERESTS

Applied cryptography, zero-knowledge proofs, verifiable computation, and privacy-preserving systems.

EDUCATION

- Sept. 2023 – **Rutgers University**, *Ph.D. student in Computer Science*, USA
Present Advised by Prof. Aarushi Goel and Prof. Xiong (Leo) Fan.
- Sept. 2019 – **Rutgers University**, *M.S. in Computer Science*, USA, GPA:3.85
May 2023
- Sept. 2012 – **Sun Yat-sen University**, *B.E. in Software Engineering*, China
Jun. 2016

MANUSCRIPTS

- [ePrint 2026] Aarushi Goel, Gabriel Kaptchuk, and **Yuange Li** (authors in alphabetical order), Sunshine Systems: Enabling Privacy-Preserving Compliance with Freedom of Information Laws, Cryptology ePrint Archive, Paper 2026/2072, 2026
ePrint: 2026/2072

PUBLICATIONS

- [IEEE Euro S&P 2026] **Yuange Li** and Xiong Fan, SUMMER: Recursive Zero-Knowledge Proofs for Scalable RNN Training, In: *IEEE European Symposium on Security and Privacy (EuroS&P)*, pp. 99–117, 2026
DOI: 10.1109/EuroSP68448.2026.00018

RESEARCH PROJECTS

Sunshine Systems: Privacy-Preserving Verifiable Document Retrieval, [Preprint, 2026]

- Developed a cryptographic framework for privacy-preserving, verifiable document retrieval over committed document collections.
- Constructed a keyword-query protocol using zero-knowledge proofs and lookup arguments to verify result completeness and correctness without revealing unrelated documents.
- Implemented and evaluated the system on low-cost hardware to assess its computational overhead and practical feasibility.

SUMMER: Recursive Zero-Knowledge Proofs for Scalable RNN Training, [IEEE Euro S&P 2026]

- Designed and implemented the first end-to-end Zero-Knowledge Proof of Training (zkPoT) system for Recurrent Neural Networks (RNNs).
- Developed a unified verification pipeline for Backpropagation Through Time (BPTT), combining sumcheck-based proofs for linear layers and LogUp-style lookup arguments for non-linear activations.
- Achieved constant proof size and verification time independent of training iterations via recursive proof composition (IVC), outperforming state-of-the-art baselines by 8.5x in proving speed and 11.6x in memory efficiency.

WORK EXPERIENCE

- May 2026 – **Chainlink Labs**, *Research Engineer Intern, mentored by Dr. Gregory Neven*
Present
 - Designing zero-knowledge proof approaches for privacy-preserving, verifiable computation.
 - Investigating privacy trade-offs in cryptographic protocol design for blockchain applications.
- Jun. 2025 – **Cysic, Inc.**, *Research Engineer Intern, mentored by Dr. Xiong (Leo) Fan*
Apr. 2026
 - Developed **SUMMER**, an end-to-end Zero-Knowledge Proof of Training (zkPoT) system for RNNs that achieved 8.5x performance improvement over baselines.
 - Explored and prototyped next-generation zero-knowledge primitives for Large Language Models (LLMs) and other advanced cryptographic methods.

- Jun. 2024 – **Cysic, Inc.**, *Research Engineer Intern*, mentored by Dr. Xiong (Leo) Fan
- Aug. 2024
- Conducted research and development focused on Halo2 and Plonky2 for cryptographic primitives in decentralized applications.
 - Designed and optimized zk-SNARK protocols to improve proof generation and verification efficiency in zero-knowledge proof systems.
 - Tested, analyzed, and applied open-source libraries, including SP1, RISC ZERO, Intmax2, EZKL, and Expander for zero-knowledge solutions.
- Aug. 2020 – **SenseTime**, *Senior Solutions Manager*, Shenzhen, China
- Aug. 2022
- Designed tailored solutions and recommended process improvements to meet current and future business needs.
 - Assessed customers' strategic needs and discovered key business requirements to recommend the company's products, solutions, and services, articulating their advantages and business value.
 - Collaborated with product teams locally and overseas to understand available products and propose solutions best suited to client needs.
 - Provided timely, prioritized, and complete customer-based feedback to Product Management, Research, Sales, Engineering and/or Support regarding client's business cases, requirements and issues.
- Aug. 2016 – **Huawei Technologies Co., Ltd.**, *Android Development Engineer*, Shenzhen, China
- Jul. 2019
- Developed Android client modules for Huawei AppGallery and Essential Apps, including multilingual support and backend API integration.
 - Optimized application performance and integrated cloud services for distributed deployment.
 - Led development of the Huawei VRStore client, including 3D user interfaces.
 - Optimized VR data streaming and evaluated cross-platform compatibility.
- Jul. 2015 – **Huawei Technologies Co., Ltd.**, *Software Engineer Intern*, Shenzhen, China
- Sept. 2015
- Maintained the Huawei AppGallery Android client, focusing on debugging, performance optimization, and multi-region deployment.

AWARDS & CERTIFICATES

2012–2013 **University Fellowship**, Sun Yat-sen University

TEACHING EXPERIENCE

- Spring 2026 **Teaching Assistant**, CS442: *Introduction to Cryptography*, Rutgers University
- Fall 2025 **Teaching Assistant**, CS344: *Design and Analysis of Computer Algorithms*, Rutgers University
- Fall 2024 **Teaching Assistant**, CS512: *Introduction to Data Structures and Algorithms*, Rutgers University
- Spring 2024 **Teaching Assistant**, CS419: *Computer Security*, Rutgers University
- Fall 2023 **Teaching Assistant**, CS206: *Introduction to Discrete Structures II*, Rutgers University
- Spring 2015 **Teaching Assistant**, *Communication Theory*, Sun Yat-sen University

ACADEMIC SERVICE

External
Reviewer

FC'24, Crypto'25, IEEE S&P'26, Crypto'26

PROGRAMMING LANGUAGES

Java, Python, C++, C, Rust